

Karnataka Board

Biology

Class 12

Instructions:

i) The question paper consists of four parts – A, B, C, and D

Part – D consists of two sections – Section I and Section II

All the Parts are compulsory

iii) Draw diagrams wherever necessary. Unlabeled diagrams or illustrations do not attract any marks.

PART - A

Answer the following questions in one word or one sentence. ($10 \times 1 = 10$)

1) Name the reproductive cycle that occurs in non-primates.

Ans: Oestrous cycle is the reproductive cycle that occurs in non-primates.

2) Name the layer of uterus that exhibits strong contraction during parturition.

Ans: Myometrium is the layer of uterus that exhibits strong contraction during parturition.

3) Which chromosome of man has least number of genes?

Ans: 'Y' chromosome of man has least number of genes.

4) Write the infectious form of plasmodium which enters body through mosquito bite.

Ans: Sporozoites is infectious form of plasmodium which enters body through mosquito bite.

5) Give an example for an internal specific hybrid animals.

Ans: Mule

6) Write the scientific name of the source organism for citric acid.

Ans: Aspergillus niger the scientific name of the source organism for citric acid.

7) Mention a gene that codes for insecticidal protein in Bt cotton.

Ans: cry gene (cryIAC, cryIIAb, cryIAb codes for insecticidal protein in Bt cotton.

8) Name the plasmid which present in agrobacterium tumefaciens.

Ans: Ti-plasmid is present in agrobacterium tumefaciens.

9) What is the main aim of Montreal protocol?

Ans: Main aim of the Montreal Protocol to control the emission of ozone depleting substances.

10) Name the type of food chain that is measured conduct for energy flow in aquatic ecosystem.

Ans: Grazing Food Chain (GFC)

PART -B

Answer any five of the following questions in 3 to 5 sentences each wherever applicable.

(5×2 = 10)

11) What is embryogenesis? Mention two important events that occurred during embryogenesis.

Ans: Embryogenesis is the process of development of embryo from the zygote The two important events that occur during embryogenesis a)Cell division b)Cell differentiation .

12) Define point mutation .Give an example for point mutation.

Ans: Mutation that arise due to change in a single base pair of a DNA Example: Sickle Cell Anaemia

13) What are the conclusions drawn by T .H. Morgan from the crossing experience in Drosophila with respect to linkage?

Ans: Conclusions drawn by T.H.Morgan from crossing experiments in Drosophila: When the genes are loosely linked show higher recombination and when the genes are tightly linked show low recombination.

14) What is an allergy? Name the two chemical released by mast cells in the body during allergy.

Ans: The exaggerated response of the immune system to certain antigens present in the environment is called allergy Histamine and Serotonin are the two chemicals secreted by the mast cells during allergy.

15) Mention the four traits for which plant breeding is done.

Ans: The four traits for which plant breeding is done:

- a) Increased tolerance to environmental stresses (Salinity, extreme temperature, drought)
- b) Resistance to pathogens (viruses, fungi, bacteria)
- c) Increased tolerance to insect pests
- d) Increased crop yield and improved quality.

16) With an example explain the convention for naming restriction endonuclease scientifically.

Ans: Convention for naming restriction endonucleases scientifically Eg: EcoRI comes from Escherichia coli RY13 First letter of the name comes from the genus Escherichia and the second letter from the species of the prokaryotic cell from which they were isolated coli.

In EcoRI, the letter 'R' is derived from the name of the strain Escherichia coli RY13. Roman number following the names indicates the order in which the enzymes were isolated from that strain of bacteria.

17) A prey develops defence against a Predator. Justify the statement with two examples in animals.

Ans: (i) Some species of insects and frogs are cryptically coloured (Camouflaged) to avoid being detected easily by the predator

ii) Monarch Butterfly is highly distasteful to its predator (bird) because of special chemical present in its body .

18) Mention any two mechanism how human body compensates low oxygen availability at higher altitude.

Ans: The human body compensate low oxygen availability at higher altitude by increasing red blood cell production, decreasing the binding affinity of haemoglobin and by increasing breathing rate.

PART - C

Answer any five of the following questions in about 40 to 80 words each wherever applicable. $5 \times 3 = 15$

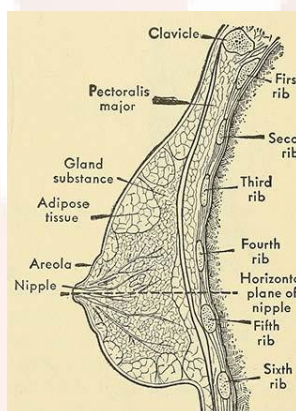
19) Write the advantages offered by the seeds to angiosperm.

Ans: Seeds offer several advantages to angiosperms. They are followings:

- Seeds have better adaptive strategies for dispersal to new habitats and help the plant species grow in a new area.
- As the seeds have sufficient reserve food the young seedlings are nourished until they are capable of photosynthesis on their own.
- The hard seed coat provides protection to the embryo.
- As they are the products of sexual reproduction they contribute new genetic combinations to new generations that leads to variations which is the necessary tools for evolution.

20) Draw a neat labelled diagram of sectional view of human mammary gland.

Ans:



Mammary Gland

21) DNA is the better genetic material than RNA justify the statement with three comparative reasons.

Ans; DNA is better genetic material than RNA due to following reasons:

- Presence of thymine at the place of uracil confers additional stability to DNA

- It provides the scope for slow changes (mutation) because RNA being unstable mutate at the faster rate.

- The 2'-OH group present at every nucleotide in RNA is reactive and makes RNA labile and degradable. But DNA is less reactive and structurally more stable.

22) Enumerate on convergent and divergent evolution with suitable examples in plants.

Ans: Convergent Evolution: It creates analogous structures that evolve for same function and hence having similarities Eg: Sweet potato (root modification) and Potato (Stem modification).

Divergent Evolution: The same structure developed along different direction due to adaptations to different need. This is divergent evolution and these structures are homologous Eg: Thorns and tendrils of Bougainvillea and Cucurbita.

23) Mention any two group of carcinogens with one example for each.

Ans: The agents which causes cancer are called carcinogens are:

- Physical carcinogens- Ionising radiations, X-rays
- Chemical carcinogens- Tobacco Smoke
- Biological Carcinogens-Oncogenic viruses

24) a. Mention four rules required for Recombinant DNA technology. 2

Ans: Tools required for recombinant DNA technology Restriction enzymes, cloning vector, competent host, polymerase enzymes, and ligases.

b. With reference to gel electrophoresis what is elution. 1

Ans: The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece is called elution.

25) Ecological pyramids have limitations justify the statement with three reasons.

Ans: Limitations of ecological pyramids

- It does not take into account the same species belonging to two or more trophic levels
- It assumes a simple food chain, it never exists in nature
- It does not accommodate food web
- Saprophytes are not given place in ecological pyramids

26) Describe three factors which affect on decomposition.

Ans: Factors which affect decomposition:

- In a particular climatic condition, decomposition rate is slower if detritus is rich in lignin and chitin and quicker, if detritus is rich in nitrogen and water soluble substances like sugars.
- Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes.
- Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in buildup of organic materials

PART - D

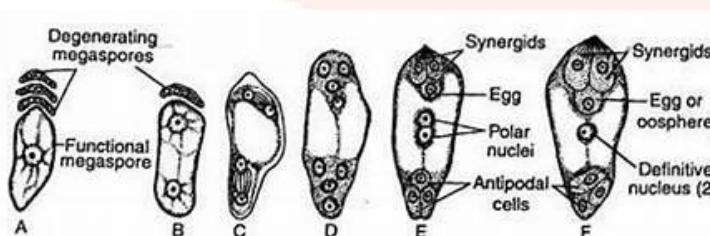
SECTION – I

Answer any four of the following questions in about 200 to 250 words each, wherever applicable 4 × 5 = 20

27) What is megasporangium? Explain the development of 8- nucleate embryo sac in flowering plants.

Ans: The process of formation of megaspores from the megaspore mother cell is called megasporogenesis. Development of eight nucleate embryo sac in flowering plants:

- The nucleus of the functional megaspore divides mitotically to form two nuclei, which moves to the opposite poles, forming the 2-nucleate embryo sac.
- Two more sequential mitotic nuclear divisions result in the formation of the 4-nucleate and later the 8-nucleate stage of the embryo sac.
- These divisions are strictly free nuclear i.e, nuclear divisions are not followed immediately by cell wall formation.
- After the 8-nucleate stage cell walls are laid down leading to the organisation of the typical female gametophyte or embryo sac.



Development of Embryo sac

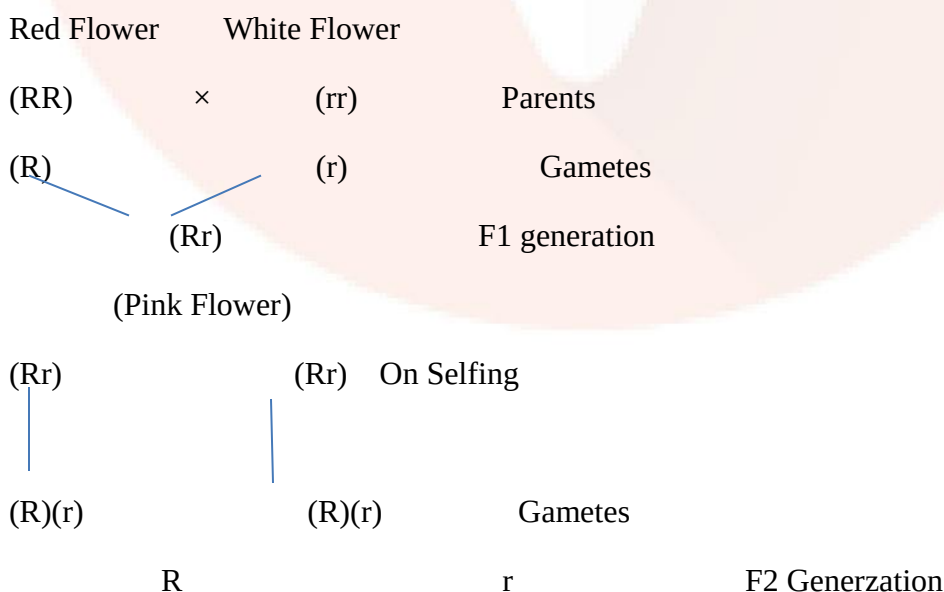
28) What are contraceptives? Explain four different non- surgical contraceptives.

Ans: Contraceptives are the devices which prevents unwanted pregnancy. Non-surgical Contraceptive Methods are followings:

- 1) Barrier Method: Ovum and sperm are prevented from physically meeting with the help of barriers such methods are available for both males and females
- 2) Intra Uterine devices (IUD's): These are inserted into the uterus through vagina. Types are: Non Medicated (Lippes Loop), Cu releasing IUDs (Cu T, Cu7) Hormone IUDs(LNG- 20) .They increase the phagocytosis of the sperm within uterus and Cu ions released suppress sperm motility and fertilising capacity of the sperms. The hormone releasing IUDs in addition make the uterus unsuitable for implantation and the cervix hostile to the sperms
- 3) Implants: Progestogens or progesterone or implants under the skin
- 4) Oral contraceptive pills: Oral administration of small doses of either progesterone or progesterone estrogen combinations in the form of pills.

29) What is incomplete dominance explain it with reference to flower colour in Snapdragon.

Ans: In heterozygous condition both the alleles fails to dominate each other and exhibit intermediate characters in F1 generation it is called as incomplete dominance This kind of inheritance is found in It is cross between true breeding red flower (RR) and true breeding white flower (rr), the F1 (Rr) was pink). When F1 was self-pollinated the F2 resulted in the following ratio 1(RR)Red:2 (Rr)Pink:1(rr)White.



RR (Red)	Rr (Pink)
Rr (Pink)	rr (White)

Phenotypic Ratio: Red : Pink: White
1 : 2 : 1

Genotypic Ratio :Rr: Rr: Rr
1 : 2 : 1

30) a. Explain the role of the organism as biocontrol agents.

Ans: Role of organisms as bio controlagents are followings:

- The ladybird and dragonflies are useful to get rid of aphids and mosquitoes
- Bacillus thuringiensis used to control butterflies caterpillars. Dried spores are mixed with water and sprayed on to plants such as Brassicas and fruit trees where these are eaten by the insect larvae. In the gut of the larvae the toxin is released and the larvae gets killed.
- Trichoderma- a free living fungus used to control several plant pathogens.

b. What is significance of BOD? 1

Ans: Significance of BOD: BOD is the measure of organic matters present in the water. Greater the BOD of waste water more is its polluting potential.

c. Give an example for a fungus found in mycorrhiza. 1

Ans: Glomus

31. Explain five benefits of creating transgenic animals.

Ans: Benefits of creating transgenic animals

i) Study of normal physiology and development:

- Useful to study gene regulation, their effect on the normal functions of the body and its development
- For example, study of complex growth factors like insulin-like growth factors.

ii) Study of disease:

- Study of genes which are responsible for diseases in human and their treatment
- Transgenic models have been developed for many human diseases like cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's disease

iii) Biological products:

- Useful biological products can be produced by introducing into transgenic animals the portion of DNA (or genes) which codes for a particular product
- For example, human protein (α -1-antitrypsin) is used to treat emphysema
- In 1997, the first transgenic cow, Rosie produced human protein-enriched milk (2.4g/L)
- The milk contained the human alpha-lactalbumin and was more nutritionally balanced for human babies than natural cow milk

iv) Vaccine safety:

- Transgenic mice are developed to test safety of vaccines before being used on humans. Eg: polio vaccine.

v) Chemical safety testing:

- Transgenic animals are made to carry genes which make them more sensitive to the toxic substances than non-transgenic animals

32) a. Tropical region have greater biodiversity than temperate region .Justify the statement with three reasons.

Ans: Tropical region has greater biodiversity than temperate region because

i)Unlike temperate regions subjected to frequent glaciation in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification .

ii) Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable, promotes niche specialisation and lead to greater species diversity.

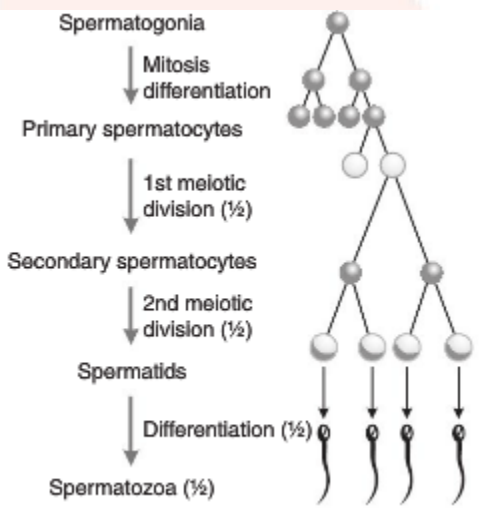
iii) There is more solar energy available in the tropics, which contribute to higher productivity

b. The use of CNG is better than petrol or diesel. Give four reasons.

Ans: Use of CNG is better than petrol or diesel because CNG burns most efficiently, unlike petrol or diesel, in the automobiles and very little of it is left unburnt. CNG is cheaper than petrol or diesel. It cannot be siphoned off by thieves and adulterated like petrol or diesel.

SECTION - II

33) a. Write the schematic representation of spermatogenesis. 3



SPERMATOGEGESIS

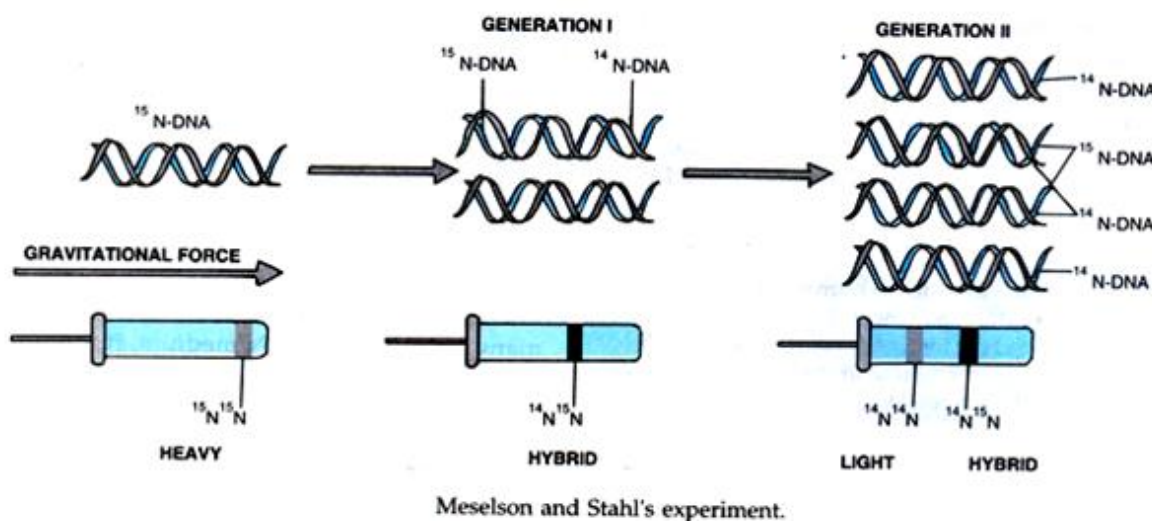
b. Write the two events that occur in the ovary and uterus during the follicular phase of menstrual cycle. 2

Ans: During follicular phase, the primary follicles in the ovary grow to become a fully mature graafian follicle under the stimulation of FSH and LH. In the uterus endometrium regenerates through proliferation under the stimulation of estrogen hormone.

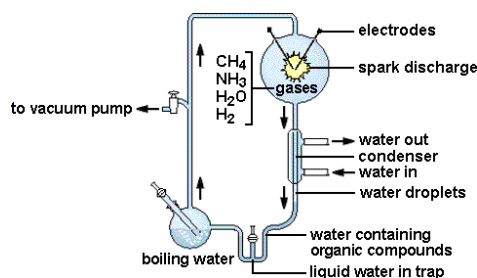
34) DNA replication is said to be semi conservative. Why? Describe the experimental proof of Meselson and Stahl to show DNA replication is semiconservative.

Ans: During DNA replication in daughter DNA molecules one strand is parental and another is newly synthesized. Hence DNA replication is called semi conservative. Semiconservative method was shown by Messelson and Stahl (1958) in *Escherichia coli*

- They grew *E. coli* in a medium containing $^{15}\text{NH}_4\text{Cl}$ (^{15}N is the heavy isotope of nitrogen) as the only nitrogen source for many generations. The result was that ^{15}N was incorporated into newly synthesised DNA. This heavy DNA molecule could be distinguished from the normal DNA by centrifugation in a Cesium chloride (CsCl) density gradient
- Then they transferred the cells into a medium with normal $^{14}\text{NH}_4\text{Cl}$ and took samples at various definite time intervals as the cells multiplied, and extracted the DNA that remained as double-stranded helices. The various samples were separated independently in CsCl gradients to measure the densities of DNA.
- Thus, the DNA that was extracted from the culture one generation after the transfer from ^{15}N to ^{14}N medium had a hybrid or intermediate density. DNA extracted from the culture after another generation was composed of equal amounts of this hybrid DNA and of light DNA.



35) a. Draw a neat and labeled diagram of miller's experiment. 3



MILLER EXPERIMENT

b. Mention two assumptions of Oparin and Haldane with reference to

‘Origin of Life’. 2

Ans: Assumption of Oparin and Haldane

- First form of life could have come from pre-existing non-living organic molecules (eg: RNA, protein).
- Formation of life was preceded by chemical evolution i.e., formation of diverse organic molecules from inorganic constituents.

36) Name the technology that can successfully increase the size of herd in a short time and explain the steps involved in the technology.

Ans: Multiple Ovulation Embryo Transfer Technology (MOET): It is one of the programmes for increasing the herd size

- It is used to improve chances of successful production of hybrids.
- Cow is administered hormones with FSH like activity induces follicular maturation and production of many numbers of ova called Super ovulation.
- Production of 6-8 eggs instead of 1 egg per oestrous cycle.
- The female is either mated with an elite bull or artificially inseminated.
- Non-surgical recovery of fertilised eggs at 8-32 celled stage.
- Each one transferred to the surrogate mother. The genetic mother is available for another round of super ovulation.

37) a. Mention four Evil Quartet that cause depletion of biodiversity.

2

a) Four 'Evil Quartet' which causes depletion of biodiversity are:

i) Habitat loss and fragmentation.

ii) Over Exploitation.

iii) Alien species invasion.

iv) Co-extinction.

b. Write two suspended activities in animals against abiotic stresses with suitable examples.

2

Ans: i) Hibernation (winter sleep): It is a suppressed metabolic state of an organism to avoid winter season Eg: Bears going into hibernation in winter.

ii) Aestivation (summer sleep): Hibernation during the summer season is called aestivation Eg: Fishes avoid summer related problems like heat and dessication.

iii) Diapause: It is the delay in the development during adverse environmental condition Eg: Many zooplanktons undergoes a stage of suspended development in unfavourable conditions.

c. Among vertebrates which group of animals has the highest number of height in global biodiversity? 1

Ans: Fishes.
